MONOIDAL VECTOR BUNDLES

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Abstract. We follow the road of Alain Connes to define noncommutative vector bundle based on the Serre-Swan theorem. We define define a monoidal vector bundle as finitely generated projective module over a monoid over a monoidal category. We analyze this monoidal vector bundles.

Introduction

Alain Connes describes in his book [4] how he started to define noncommutative topology starting with the duality of Gelfand that states that there is a duality between the category of compact Hausdorff spaces and unital commutative C^*-algebras. So he defines the category of noncommutative compact Hausdorff spaces as the opposite category of the category of unital C^*-algebras. This point of view has several advantages, but we will focus in one, the one that is important from a philosophical point of view to this paper. It is possible to continue with this duality thanks to the Serre-Swan theorem that sates that there is an equivalence of categories between the category of complex bundles of a compact space X and the category of finitely generated projective modules over the ring of continue complex valued functions over X, $C(X)$. So Connes defines a virtual vector bundle as a finitely generated porjective module over...

[2] [6]

1. Monoidal Categories

Definition 1.1. *A **monoidal category** $(\mathcal{C}, \otimes, \alpha, e, \iota)$ consists of:*

(1) *A bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ called tensor.*

(2) *A natural isomorphism $\alpha : (_ \otimes _) \otimes _ \to _ \otimes (_ \otimes _)$ called **associativity constraint** which satisfies the **pentagon axiom**:*

$$\begin{tikzcd}
 & (W \otimes X) \otimes (Y \otimes Z) \arrow[dr, "\alpha_{W,X,Y\otimes Z}"] & \\
((W \otimes X) \otimes Y) \otimes Z \arrow[ur, "\alpha_{W\otimes X,Y,Z}"] \arrow[d, "\alpha_{W,X,Y}\otimes 1_Z"'] & & W \otimes (X \otimes (Y \otimes Z)) \\
(W \otimes (X \otimes Y)) \otimes Z \arrow[rr, "\alpha_{W,X\otimes Y,Z}"'] & & W \otimes ((X \otimes Y) \otimes Z) \arrow[u, "1_W \otimes \alpha_{X,Y,Z}"']
\end{tikzcd}$$

(3) *A distinguished object $e \in \mathcal{C}$ together with natural isomorphisms $\lambda_X : X \to e \otimes X$, $\rho_X : X \to X \times e$ called **unit constraints** which satsify the **triangle axiom**:*

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$$\begin{tikzcd}
(X \otimes e) \otimes Y \arrow[rr, "\alpha_{X,e,Y}"] & & X \otimes (e \otimes Y) \\
& X \otimes Y \arrow[ul, "\rho_A \otimes 1_Y"] \arrow[ur, "1_X \otimes \lambda_Y"'] &
\end{tikzcd}$$

Furthermore, $\lambda_e = \rho_e$ and the following diagrams commute:

$$\begin{tikzcd}
(e \otimes X) \otimes Y \arrow[rr, "\alpha_{e,X,Y}"] & & e \otimes (X \otimes Y) \\
& X \otimes Y \arrow[ul, "\lambda_X \otimes 1_Y"] \arrow[ur, "\lambda_{X \otimes Y}"'] &
\end{tikzcd}$$

$$\begin{tikzcd}
(X \otimes Y) \otimes e \arrow[rr, "\alpha_{e,X,Y}"] & & X \otimes (Y \otimes e) \\
& X \otimes Y \arrow[ul, "\rho_{X \otimes Y}"] \arrow[ur, "1_X \otimes \rho_Y"'] &
\end{tikzcd}$$

Definition 1.2. *Let $L : \mathcal{C} \to \mathcal{D}$ and $R : \mathcal{D} \to \mathcal{C}$ be a pair of functors. L is left adjoint to the functor R, if there exist natural transformations $\eta : 1_{\mathcal{C}} \to R \circ L$, $\varepsilon : L \circ R \to 1_D$ such that the following diagrams commute:*

$$\begin{tikzcd}
L \arrow[r, "L \circ \eta"] \arrow[dr, "1_L"'] & L \circ R \circ L \arrow[d, "\varepsilon \circ L"] \\
& L
\end{tikzcd}
\qquad
\begin{tikzcd}
R \arrow[r, "\eta \circ R"] \arrow[dr, "1_R"'] & R \circ L \circ R \arrow[d, "R \circ \varepsilon"] \\
& R
\end{tikzcd}$$

Is easy to see that if a category has binary products and terminal object, then is a monoidal category with tensor the product and unit the terminal object.

Definition 1.3. *A cartesian monoidal category is a category whose monoidal structure is given by product and the terminal object.*

This give us several examples of monoidal categories as $\mathcal{S}$ the category of sets and functions, and $\mathcal{T}$ the category of topological spaces and continuous functions. One particular example is the case of a bounded lattice L seen as a category, the product correspond to the infimum $\wedge$ and the unit to the greatest element 1. Also, we have the category of abelian groups $\mathcal{A}$ with group morphisms, this category is cartesian monoidal category but in general we will assume that its monoidal structure is given by tensor product over the integers $\otimes_{\mathbb{Z}}$ and unit $\mathbb{Z}$. A similar situation happens to the category of K-vector space $\mathcal{V}_K$ and K-linear transformations over a field K, again, even if it is a cartesian monoidal category, we will asume that is monoidal structure is given by the tensor product over the field K, $\otimes_K$, and the unit is K.

*Natural transformations above mentioned are known as **unity** and **counity** of the adjunction respectively. And the pair of commutative diagrams are known as **triangular identities**.*

Definition 1.4. *A **monoid** in a monoidal category $(\mathcal{C}, \otimes, \alpha, e, \iota)$ is a triple (A, μ, η) where $\mu : A \otimes A \to A$, $\eta : e \to A$ are morphisms in $\mathcal{C}$ called **product** and **unit** such that the following diagrams commute:*

$$\begin{tikzcd}
(A \otimes A) \otimes A \arrow[r, "\alpha_{A,A,A}"] \arrow[d, "\mu \otimes 1_A"'] & A \otimes (A \otimes A) \arrow[r, "1_A \otimes \mu"] & A \otimes A \arrow[d, "\mu"] \\
A \otimes (A \otimes A) \arrow[rr, "\mu"'] & & A
\end{tikzcd}$$

$$\begin{tikzcd}
e \otimes A \arrow[r, "\eta \otimes 1_A"] \arrow[dr, "\lambda_A^{-1}"'] & A \otimes A \arrow[d, "\mu"] & A \otimes e \arrow[l, "1_A \otimes \eta"'] \arrow[dl, "\rho_A^{-1}"] \\
& A &
\end{tikzcd}$$

Definition 1.5. *Let (A, μ, η) be a monoid in a monoidal category $(\mathcal{C}, \otimes, \alpha, e, \iota)$. A left $A-$module is a pair (M, ρ) where M is an object in $\mathcal{C}$ and $\rho : A \otimes M \to M$ is a morphism in $\mathcal{C}$, such that the following diagrams commute:*

$$\begin{tikzcd}
(A \otimes A) \otimes A \arrow[r, "\alpha_{A,A,M}"] \arrow[d, "\mu \otimes 1_M"'] & A \otimes (A \otimes M) \arrow[d, "1_A \otimes \rho"] \\
A \otimes M \arrow[r, "\rho"'] & M & A \otimes M \arrow[l, "\rho"]
\end{tikzcd}$$

$$\begin{tikzcd}
e \otimes M \arrow[r, "\lambda_M^{-1}"] \arrow[d, "\eta \otimes 1_M"'] & M \arrow[d, "1_M"] \\
A \otimes M \arrow[r, "\rho"'] & M
\end{tikzcd}$$

Definition 1.6. *Let (A, μ, η) be a monoid in a monoidal category $(\mathcal{C}, \otimes, \alpha, e, \iota)$ and consider a pair (M, ρ), (N, σ) of left $A-$modules. A morphism $f : M \to N$ in $\mathcal{C}$ is a morphism of $A-$modules if the following diagram commutes:*

$$\begin{tikzcd}
A \otimes M \arrow[r, "1_A \otimes f"] \arrow[d, "\rho"'] & A \otimes N \arrow[d, "\sigma"] \\
M \arrow[r, "f"'] & N
\end{tikzcd}$$

For reference about monoidal categories, we suggest [1] and [5].

2. Modules over a Monoid

Proposition 2.1. *Let (A, μ, η) be a monoid in a monoidal category $(\mathcal{C}, \otimes, \alpha, e, \iota)$. The collection of all left $A-$modules and composition of morphisms in $\mathcal{C}$ give rise a category, such category is called* A-Mod.

Proposition 2.2. *Let (A, μ, η) be a monoid in a monoidal category $(\mathcal{C}, \otimes, \alpha, e, \iota)$. Consider the following assignments:*

- *On objects. Let X be an object in $\mathcal{C}$, we define*
$$F(X) := (A \otimes X, \rho)$$
where $\rho = \mu \otimes 1_X \circ \alpha^{-1}_{A,A,X} : A \otimes (A \otimes X) \to A \otimes X$.

- *On morphisms. Let $f : X \to Y$ be a morphism in $\mathcal{C}$, we define*
$$F(f) := 1_A \otimes f : A \otimes M \to A \otimes N.$$

We claim that $F : \mathcal{C} \to$ A-Mod *is a covariant functor.*

Proof. We start by showing that indeed $F(X) = (A \otimes X, \rho)$ is a left A-module for all $X \in \mathcal{C}$, where $\rho = \mu \otimes 1_X \circ \alpha^{-1}_{A,A,X} : A \otimes (A \otimes X) \to A \otimes X$.

It is necessary to prove that the following diagrams commute:

$$\begin{tikzcd}
(A\otimes A)\otimes(A\otimes X) \arrow[rr, "\alpha_{A,A,A\otimes X}"] \arrow[d, "\mu\otimes 1_{A\otimes X}"'] & & A\otimes(A\otimes(A\otimes X)) \arrow[d, "1_A\otimes\rho"] \\
A\otimes(A\otimes X) \arrow[r, "\rho"'] & A\otimes X & A\otimes(A\otimes X) \arrow[l, "\rho"]
\end{tikzcd}$$

$$\begin{tikzcd}
e\otimes(A\otimes X) \arrow[r, "\lambda^{-1}_{A\otimes X}"] \arrow[d, "\eta\otimes 1_{A\otimes X}"'] & A\otimes X \arrow[d, "1_{A\otimes X}"] \\
A\otimes(A\otimes X) \arrow[r, "\rho"'] & A\otimes X
\end{tikzcd}$$

Since A is a monoid, it is satisfied that $\mu\circ\mu\otimes 1_A = \mu\circ 1_A\otimes\mu\circ\alpha_{A,A,A}$. Now consider the morphism $1_X : X \to X$ and applying the bifunctor $\otimes$ we get the following:

$$\mu\otimes 1_X\circ(\mu\otimes 1_A)\otimes 1_X = \mu\otimes 1_X\circ(1_A\otimes\mu)\otimes 1_X\circ\alpha_{A,A,A}\otimes 1_X. \tag{1}$$

Let's start working with the left side of the equation. We have the next sutuation:

$$\begin{tikzcd}
((A\otimes A)\otimes A)\otimes X \arrow[r, "\alpha_{A\otimes A,A,A}"] \arrow[d, "(\mu\otimes 1_A)\otimes 1_X"'] & (A\otimes A)\otimes(A\otimes X) \arrow[d, "\mu\otimes 1_{A\otimes X}"] \\
(A\otimes A)\otimes X \arrow[r, "\alpha_{A,A,X}"] \arrow[d, "\mu\otimes 1_X"'] & A\otimes(A\otimes X) \arrow[d, "\rho"] \\
A\otimes X \arrow[r, "1_{A\otimes X}"] & A\otimes X
\end{tikzcd}$$

since $\alpha : (_\otimes_)\otimes_ \to _\otimes(_\otimes_)$ is natural in each variable the upper square commutes so the whole diagram commutes, thus

$$\rho\circ\mu\otimes 1_{A\otimes X} = (\mu\otimes 1_X\circ(\mu\otimes 1_A)\otimes 1_X)\circ\alpha^{-1}_{A\otimes A,A,X}. \tag{2}$$

Now we continue working with the right side of the equation (1), the following diagram illustrates the situation:

$$\begin{tikzcd}
((A\otimes A)\otimes A)\otimes X \arrow[rrr, "\alpha_{A\otimes A,A,X}"] \arrow[d, "\alpha_{A,A,A}\otimes 1_X"'] & & & (A\otimes A)\otimes(A\otimes X) \arrow[d, "\alpha_{A,A,A\otimes X}"] \\
(A\otimes(A\otimes A))\otimes X \arrow[r, "\alpha_{A,A\otimes A,X}"] \arrow[d, "(1_A\otimes\mu)\otimes 1_X"'] & A\otimes A\otimes((A\otimes A)\otimes X) \arrow[r, "1_A\otimes\alpha_{A,A,X}"] & & A(A\otimes(A\otimes X)) \arrow[d, "1_A\otimes(\mu\otimes 1_X\circ\alpha^{-1}_{A,A,X})"] \\
(A\otimes A)\otimes X \arrow[rrr, "\alpha_{A,A,X}"] \arrow[d, "\mu\otimes 1_X"'] & & & A\otimes(A\otimes X) \arrow[d, "\mu\otimes 1_X\circ\alpha^{-1}_{A,A,X}"] \\
A\otimes X \arrow[rrr, "1_{A\otimes X}"] & & & A\otimes X
\end{tikzcd}$$

by the pentagon axiom the upper square commutes and once again since $\alpha : (_\otimes_)\otimes_ \to _\otimes(_\otimes_)$ is natural in each variable the middle square commutes so the whole diagram commutes, thus

$$(\mu\otimes 1_X\circ(1_A\otimes\mu)\otimes 1_X\circ\alpha_{A,A,A}\otimes 1_X)\circ\alpha^{-1}_{A\otimes A,A,A} = \rho\circ 1_A\otimes\rho\circ\alpha_{A,A,A\otimes X}. \tag{3}$$

By equations (1), (2) and (3) we conclude that

$$\rho \circ \mu \otimes 1_{A\otimes X} = \rho \circ 1_A \otimes \rho \circ \alpha_{A,A,A\otimes X}$$

as we wanted. It remains to show that (2) commutes.

Since A is a monoid, it is satisfied that $\mu \circ \eta \otimes 1_A = \lambda_A^{-1}$. Now consider the morphism $1_X : X \to X$ and appliying the bifunctor $\otimes$ we get the following:

$$\mu \otimes 1_X \circ (\eta \otimes 1_A) \otimes 1_X = \lambda_A^{-1} \otimes 1_X. \tag{4}$$

Once again, since $\alpha : (_ \otimes _) \otimes _ \to _ \otimes (_ \otimes _)$ is natural in each variable the following diagram commutes:

$$\begin{tikzcd}
(e \otimes A) \otimes X \arrow[r, "\alpha_{e,A,X}"] \arrow[d, "(\eta\otimes 1_A)\otimes 1_X"'] & e \otimes (A \otimes X) \arrow[d, "\eta\otimes 1_{A\otimes X}"] \\
(A \otimes A) \otimes X \arrow[r, "\alpha_{A,A,X}"] \arrow[d, "\mu\otimes 1_X"'] & A \otimes (A \otimes X) \arrow[d, "\rho"] \\
A \otimes X \arrow[r, "1_{A\otimes X}"] & A \otimes X
\end{tikzcd}$$

therefore

$$\rho \circ \eta \otimes 1_{A\otimes X} \circ \alpha_{e,A,X} = \mu \otimes 1_X \circ (\eta \otimes 1_A) \otimes 1_X \tag{5}$$

notice that

$$\lambda^{-1} \otimes 1_X \circ \alpha_{e,A,X}^{-1} = \lambda_{A\otimes X}^{-1} \tag{6}$$

by the identities of the definition of monoidal category. By (5) and (6) we conclude that

$$\rho \circ \eta \otimes 1_{A\otimes X} = \lambda_{A\otimes X}^{-1}$$

as we wanted, therefore $F(X) = (A \otimes X, \rho)$ is a left A-module for all $X \in \mathcal{C}$.

Now we study the assignment on morphisms. Let $f : X \to Y$ be a morphism in $\mathcal{C}$, consider the next situation:

$$\begin{tikzcd}
A \otimes (A \otimes X) \arrow[r, "1_A\otimes(1_A\otimes f)"] \arrow[d, "\alpha_{A,A,X}^{-1}"'] & A \otimes (A \otimes Y) \arrow[d, "\alpha_{A,A,Y}^{-1}"] \\
(A \otimes A) \otimes X \arrow[r, "1_{A\otimes A}\otimes f"] \arrow[d, "\mu\otimes 1_X"'] & (A \otimes A) \otimes Y \arrow[d, "\mu\otimes 1_Y"] \\
A \otimes X \arrow[r, "1_A\otimes f"'] & A \otimes Y
\end{tikzcd}$$

given that $\alpha : (_ \otimes _) \otimes _ \to _ \otimes (_ \otimes _)$ is natural in each variable the upper square commutes so the whole diagram commutes, therefore

$$\begin{aligned}1_A \otimes f \circ \rho =& 1_A \otimes f \circ \mu \otimes 1_X \circ \alpha^{-1}_{A,A,X}\\ =& \mu \otimes 1_Y \circ \alpha^{-1}_{A,A,Y} \circ 1_A \otimes (1_A \otimes f)\\ =& \varrho \circ 1_A \otimes F(f)\end{aligned}$$

we conclude that $F(f) : F(X) \to F(Y)$ is a morphism of left A-modules for all $f : X \to Y \in \mathcal{C}$.

Now we check funtoriality. Let $f : X \to Y$, $g : Y \to Z$ be a morphisms in $\mathcal{C}$, given that $\otimes$ is a bifunctor it follows that:

$$\begin{aligned}F(g \circ f) =& 1_A \otimes (g \circ f)\\ =& (1_A \circ 1_A) \otimes (g \circ f)\\ =& (1_A \otimes g) \circ (1_A \otimes f)\\ =& F(g) \circ F(f).\end{aligned}$$

To conclude, consider $X \in \mathcal{C}$, given that $\otimes$ is a bifunctor it follows that:

$$F(1_X) = 1_A \otimes 1_X = 1_{A \otimes X} = 1_{F(X)}$$

therefore $F : \mathcal{C} \to$ A-Mod is a covariant functor. □

Lemma 2.1. *Let (A, μ, η) be a monoid in a monoidal category $(\mathcal{C}, \otimes, \alpha, e, \iota)$. If $(M, \varrho) \in$ A-Mod then $\varrho : A \otimes M \to M$ is a morphism in* A-Mod.

Proof. Given that M is a left A-module the following diagram commutes:

$$\begin{tikzcd}(A \otimes A) \otimes A \arrow[r, "\alpha_{A,A,M}"] \arrow[d, "\mu \otimes 1_M"'] & A \otimes (A \otimes M) \arrow[d, "1_A \otimes \varrho"] \\ A \otimes M \arrow[r, "\varrho"'] & M & A \otimes M \arrow[l, "\varrho"]\end{tikzcd}$$

then $\varrho \circ \mu \otimes 1_M \circ \alpha^{-1}_{A,A,M} = \varrho \circ 1_A \otimes \varrho$, consequently the following diagram commutes:

$$\begin{tikzcd}A \otimes (A \otimes M) \arrow[r, "1_A \otimes \varrho"] \arrow[d, "\rho"'] & A \otimes M \arrow[d, "\varrho"] \\ A \otimes M \arrow[r, "\varrho"'] & M\end{tikzcd}$$

therefore $\varrho : A \otimes M \to M$ is a morphism of left A-modules. □

Proposition 2.3. *Let (A, μ, η) be a monoid in a monoidal category $(\mathcal{C}, \otimes, \alpha, e, \iota)$. The functors*

$$\mathcal{C} \underset{U}{\overset{F}{\rightleftarrows}} \text{A-Mod}$$

define an adjunction, where F is the left adjoint and U denotes the forgetful functor.

Proof. We will prove the adjunction via triangular identities, but first we define the candidates for unity and counity.

Let (M, ϱ) be a left A-module. For counity we search for a morphism $\varepsilon_M : A \otimes M \to M$ in A-Mod, intuitively we define $\varepsilon_M := \varrho$. By lemma (2.1) we know that $\varepsilon_M = \varrho$ is a morphism in A-Mod. It remains to show that $\varepsilon : FU \to 1_{\text{A-Mod}}$ is natural.

Let $f : (X, \tau) \to (Y, \upsilon)$ be a morphism in A-Mod, it follows that next diagram commutes:

$$\begin{tikzcd}
A \otimes U(X) \arrow[r, "FU(f)=1_A\otimes U(f)"] \arrow[d, "\varepsilon_X=\tau"'] & A \otimes U(Y) \arrow[d, "\varepsilon_Y=\upsilon"] \\
X \arrow[r, "f"'] & Y
\end{tikzcd}$$

hence $\varepsilon : FU \to 1_{\text{A-Mod}}$ is a natural transformation.

Let X be an object in $\mathcal{C}$. For unity we search for a morphism $\eta_X : X \to A \otimes X$ in $\mathcal{C}$, intuitively we define $\eta_X := \eta \otimes 1_X \circ \lambda_X$. Let's check that $\eta : 1_{\mathcal{C}} \to UF$ is natural.

Let $f : X \to Y$ be a morphism in $\mathcal{C}$, since $\lambda : 1_{\mathcal{C}} \to e \otimes _$ is natural, the following diagram commutes:

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "\lambda_X"'] & Y \arrow[d, "\lambda_Y"] \\
e \otimes X \arrow[r, "1_e\otimes f"] \arrow[d, "\eta\otimes 1_X"'] & e \otimes Y \arrow[d, "\eta\otimes 1_Y"] \\
A \otimes X \arrow[r, "1_A\otimes f"] & A \otimes Y
\end{tikzcd}$$

then

$$\begin{aligned}
1_A \otimes f \circ \eta_X =& 1_A \otimes f \circ (\eta \otimes 1_X \circ \lambda_X) \\
=& (\eta \otimes 1_Y \circ \lambda_Y) \circ f \\
=& \eta_Y \circ f
\end{aligned}$$

therefore $\eta : 1_{\mathcal{C}} \to UF$ is a natural transformation. It remains to verify triangular identities.

Let X be an object in $\mathcal{C}$, we have the next situation:

$$\begin{tikzcd}
A \otimes X \arrow[r, "\lambda_X"] \arrow[d, "\rho_A\otimes 1_X"'] & A \otimes (e \otimes X) \arrow[r, "1_A\otimes(\eta\otimes 1_X)"] \arrow[d, "\alpha^{-1}_{A,e,X}"] & A \otimes (A \otimes X) \arrow[d, "\alpha^{-1}_{A,A,X}"] \\
(A \otimes e) \otimes X \arrow[r, "1_{(A\otimes e)\otimes X}"'] & (A \otimes e) \otimes M \arrow[r, "(1_A\otimes\eta)\otimes 1_X"] \arrow[d, "\rho_A^{-1}\otimes 1_X"] & (A \otimes A) \otimes X \arrow[d, "\mu\otimes 1_X"] \\
 & A \otimes X \arrow[r, "1_{A\otimes X}"'] & A \otimes X
\end{tikzcd}$$

the upper left square commutes by the identities given in the definition of monoidal category, the upper right square commutes by naturality of $\alpha : (_\otimes_)\otimes_ \to _\otimes(_\otimes_)$ and the the bottom right square commutes because A is a monoid, therefore the whole diagram commutes, it follows that:

$$\begin{aligned}
\varepsilon_{F(X)} \circ F(\eta_X) =& \varepsilon_{A\otimes X} \circ 1_A \otimes (\eta_X) \\
=& (\mu \otimes 1_X \circ \alpha^{-1}_{A,A,X}) \circ (1_A \otimes (\eta \otimes 1_X) \circ 1_A \otimes \lambda_X) \\
=& 1_{A\otimes X} \circ \rho_A^{-1} \otimes 1_X \circ \rho_A \otimes 1_X \\
=& 1_A \otimes 1_X \\
=& 1_{A\otimes X} \\
=& 1_{F(X)}
\end{aligned}$$

we have shown that next diagram commutes:

$$\begin{tikzcd}
F \arrow[r, "F\eta"] \arrow[dr, "1_F"'] & FUF \arrow[d, "\varepsilon F"] \\
& F
\end{tikzcd}$$

Let (M, ϱ) be left A-module, thus the next diagram commutes:

$$\begin{tikzcd}
e \otimes M \arrow[r, "\lambda_M^{-1}"] \arrow[d, "\eta\otimes 1_M"'] & M \arrow[d, "1_M"] \\
A \otimes M \arrow[r, "\varrho"'] & M
\end{tikzcd}$$

therefore

$$\begin{aligned}
U(\varepsilon_M) \circ \eta_{U(M)} =& \varrho \circ (\eta \otimes 1_M \circ \lambda_M) \\
=& 1_M \\
=& 1_{U(M)}
\end{aligned}$$

we have shown that next diagram commutes:

$$\begin{tikzcd}
U \arrow[r, "\eta U"] \arrow[dr, "1_U"'] & UFU \arrow[d, "U\varepsilon"] \\
& U
\end{tikzcd}$$

we conclude that $(F, U) : \mathcal{C} \to$ A-Mod is an adjoint pair. □

Proposition 2.4. *The counit ϵ is an epimorphism.*

Proof. It follows from the fact that U is faithfull. □

Proposition 2.5. *Let $\mathcal{C}$ a monoidal category with finite coproducts such that the tensor commutes with finite coproducts. If A is monoid over $\mathcal{C}$ then the category of left A-modules has finite coproducts.*

Proof. □

3. Projective Objects

Definition 3.1. *Let $L : \mathcal{C} \to \mathcal{D}$ and $R : \mathcal{D} \to \mathcal{C}$ be a pair of functors such that L is left adjoint of R. An object $P \in \mathcal{D}$ is $LR-$**projective** if there exists $f : P \to LR(P)$ such that the following diagram commute:*

$$\begin{tikzcd} & P \arrow[dl, dashed, "f"'] \arrow[d, "1_P"] \\ LR(P) \arrow[r, "\varepsilon_P"'] & P \end{tikzcd}$$

Proposition 3.1. *Let $(L, R) : \mathcal{C} \to \mathcal{D}$ be an adjoint pair, then $L(X)$ is $LR-$projective for every $X \in \mathcal{C}$.*

Proof. It follows from triangular identities. □

Proposition 3.2. *Let $(L, R) : \mathcal{C} \to \mathcal{D}$ be an adjoint pair and $\{P_i\}_{i=1}^n$ a family of $LR-$projective objects, if $P = \amalg_{i=1}^n P_i$ exists, then P is $LR-$projetive.*

Proof. We look for an arrow $f : \amalg_{i=1}^n P_i \to LR(\amalg_{i=1}^n P_i)$ such that $\varepsilon_P f = 1_P$, by universal property applied to $\amalg_{i=1}^n P_i$ we get an unique arrow $f_1 : \amalg_{i=1}^n P_i \to \amalg_{i=1}^n LR(P_i)$, such that the following diagram commutes:

$$\begin{tikzcd} P_i \arrow[r, "\mu_i"] \arrow[d, "f_i"'] & \amalg_{i=1}^n P_i \arrow[d, "f_1"] \\ LR(P_i) \arrow[r, "\tilde{\mu_i}"'] & \amalg_{i=1}^n LR(P_i) \end{tikzcd}$$

now, by universal property applied to $\amalg_{i=1}^n LR(P_i)$ we get an unique arrow $f_2 : \amalg_{i=1}^n LR(P_i) \to LR(\amalg_{i=1}^n P_i)$, such that the following diagram commutes:

$$\begin{tikzcd} LR(P_i) \arrow[r, "\tilde{\mu_i}"] \arrow[dr, "LR(\mu_i)"'] & \amalg_{i=1}^n LR(P_i) \arrow[d, "f_2"] \\ & LR(\amalg_{i=1}^n P_i) \end{tikzcd}$$

intuitively, we define $f := f_2 f_1 : \amalg_{i=1}^n P_i \to LR(\amalg_{i=1}^n P_i)$, it remains to show that $\varepsilon_P f = 1_P$. Notice that

$$\begin{aligned} \varepsilon_P f \mu_i =& \varepsilon_{\amalg P_i}(f_2 f_1)\mu_i \\ =& \varepsilon_{\amalg P_i} f_2(\tilde{\mu_i} f_i) \\ =& \varepsilon_{\amalg P_i} LR(\mu_i) f_i \\ =& \mu_i \varepsilon_{P_i} f_i \\ =& \mu_i \end{aligned}$$

fourth equality follows from naturality of $\varepsilon : LR \to 1_D$ and fifth equality follows from the fact that P_i is $LR-$projective. Therefore $\varepsilon_{\amalg P_i} f = 1_{\amalg P_i}$ by an argument of uniqueness, we conclude that $\amalg_{i=1}^n P_i$ is $LR-$projetive. □

Proposition 3.3. *Let $(L, R) : \mathcal{C} \to \mathcal{D}$ be an adjoint pair and A an initial object of $\mathcal{D}$, then A is $LR-$projective.*

Proof. Since A is an initial object there exist an unique arrow $f : A \to LR(A)$, then $\varepsilon_A f$ is an arrow from A to A, an by uniqueness it follows that $\varepsilon_A f = 1_A$, therefore A is $LR-$projective. □

Proposition 3.4. *Let $(L, R) : \mathcal{C} \to \mathcal{D}$ be an adjoint pair and $\pi : P \to M$ an split epimorphism in $\mathcal{D}$ such that P is $LR-$projective, then M is $LR-$projective.*

Proof. By hypothesis, there exists morphisms $f_1 : M \to P$, $f_2 : P \to LR(P)$ in $\mathcal{D}$, such that $\pi f_1 = 1_M$ and $\varepsilon_P f_2 = 1_P$. Notice that $LR(\pi) f_2 f_1$ is an arrow from M to $LR(M)$ and

$$\begin{aligned}\varepsilon_M(LR(\pi) f_2 f_1) =& \pi \varepsilon_P f_2 f_1 \\ =& \pi f_1 \\ =& 1_M\end{aligned}$$

therefore M is $LR-$projective. □

Definition 3.2. *Let $L : \mathcal{C} \to \mathcal{D}$ and $R : \mathcal{D} \to \mathcal{C}$ be a pair of functors such that L is left adjoint of R. An object $P \in \mathcal{D}$ satisfies the* ***lifting property*** *if for every morphisms $f : P \to N$, $g : M \to N$ in $\mathcal{D}$ such that $R(g) : R(M) \to R(N)$ is an split epimorphism, then there exists a morphism $\tilde{f} : P \to M$ in $\mathcal{D}$ such that $g\tilde{f} = f$.*

Proposition 3.5. *Let $(L, R) : \mathcal{C} \to \mathcal{D}$ be an adjoint pair and $P \in \mathcal{D}$, the following conditions are equivalent:*

1. *P is $LR-$projective.*
2. *P satisfies the lifting property.*
3. *There exists $A \in \mathcal{D}$ such that P is a retract of $LR(A)$.*

Proof. Weibel. □

4. Free Objects

Definition 4.1. *Let $(F, U) \colon \mathcal{C} \longrightarrow \mathcal{D}$ be an adjoint pair. We say that an object $X \in \mathcal{D}$ is free, if there is A in $\mathcal{C}$ such that $F(A)$ is isomorphic to X. In the case that $\mathcal{C}$ is a monoidal category, A is monoid over $\mathcal{C}$, $\mathcal{D}$ is the category of left A-modules, and the adjunction is the one that we develop in last section, we call monoidal free to a left A-module M that is free under this circumstances.*

Example 4.1. *Let $\mathcal{A}$ be the category of abelian groups with the usual tensor $\otimes_{\mathbb{Z}}$. Then a monoid over $\mathcal{A}$ is a ring with one. Morever, if A is a monoid then the concept of a left module over the monoid A matches the concept of a left modules over a the ring A. So we that $A = \mathbb{Z}_{p^2}$ with p a prime, and we notice that:*

$$\mathbb{Z}_{p^2} \otimes \mathbb{Z}_p = \mathbb{Z}_p$$

This means that $\mathbb{Z}_p$ is monoidal free. Also we notice that this does not match with classical definition of free left module, since $\mathbb{Z}_p \cong \mathbb{Z}_{p^2}^{(X)}$ for any set X just by a cardinality argument. We remark that the classical definition is the definition for the adjuntion given between the category of sets and left A-modules.

There is a difference between the projective of monoidal free modules and monoidal projective modules in terms of the counit. Since in analogy of the monoidal projective modules we might expect to that for module M to be monoidal

free to be equivalent to ϵ_M to be an isomorphism, this sentence does not hold. Immediately, if ϵ_M is an isomorphism, we have that M is free. But let take a look to $\mathcal{V}_K$ with K a field. Let A be a K-algebra and V a K-vector space. Then for the counit $\epsilon_V\colon A \otimes_K V \longrightarrow V$ is hard to be an isomorphism by an argument of dimensions. The K-dimension of $A \otimes_K V$ is the product of the dimensions, and being an isomorphism give us that $A \otimes_K V$ and V have the same dimension. So the cases when this happens is that A has dimension 1, and from this $A = K$, and the second case is when $V = 0$.

Let $\mathcal{A}$ be the category of abelian groups, and let R be a monoid over $\mathcal{A}$ which means that R is a ring. We want to classify all the rings A such that the counit $\epsilon_M\colon R \otimes_{\mathbb{Z}} M \longrightarrow M$ is an isomorphism for every left R-module M. In those rings, all the modules are monoidal free.

Let M be a left R-module. If the canonical map $\rho\colon R \times M \longrightarrow R \otimes_{\mathbb{Z}} M$ is R balance then we get that ϵ_M is an isomorphism. We recall that $\mathbb{Z}$ is an initial object in the category of rings. So for any commutative ring R we have a unique ring morphism $\xi_R\colon \mathbb{Z} \longrightarrow R$. The dominion of R, $d(R)$, is defined as the elements of $x \in R$ such that $r \otimes 1 = 1 \otimes r \in R \otimes_{\mathbb{Z}} R$. Following Stenstrom [7], the previous concept of dominion of R corresponds to the dominion of ξ_R. Also, if the ring R is commutative the definition given of dominion coincides with the definition of core of a commutative ring given by Bousfield and Kan in [3]. A commutative ring R is called solid, if $d(R) = R$. If we review the propositions of [7], those extend the ones presented in [3].

Corollary 4.1. *Let R be a commutative ring. The following are equivalent:*

(1) *ξ_R is an epimorphism of rings.*
(2) *$\mu_R\colon R \otimes_{\mathbb{Z}} R \longrightarrow R$ is an isomorphism.*
(3) *The forgetful functor U from R-Mod to $\mathcal{A}$ is full*
(4) *ϵ_M is an isomorphism for every R-module M.*

From this corollary we get the next corollary.

Corollary 4.2. *Let R be a commuative ring. If R is a solid ring, then M is monoidal free for each R-module M.*

The work of Bousfiled and Kan is very extensive so they have been classified all the solid rings.

However, a commutative ring in which every module is monoidal free does not have to be a solid ring.

The following proposition characterizes the monoidal free R-modules.

Proposition 4.1. *Let R be a ring and M be a left R-module. Then M is monoidal free if and only if there is N subgroup of M such that $RN = M$.*

Proof. $\Rightarrow$) Let K be an abelian group and $\phi\colon R \otimes_{\mathbb{Z}} K \longrightarrow M$ be an R-isomorphism. Then we put $N = \{\phi(1 \otimes x) \in M \mid x \in K\}$. We affirm that N is a subgroup of M.

Let $\phi(1 \times x), \phi(1 \times y) \in N$ with $x, y \in K$. Then

$$\phi(1 \times x) + \phi(1 \times y) = \phi(1 \times x + 1 \times y) = \phi(1 \times (x + y)) \in N$$

since $x + y \in N$. Therefore N is a subgroup of M. Now, we affirm that $RN = M$. Let $y \in M$. Then there is $x \in R \otimes_{\mathbb{Z}} K$ with $\phi(x) = y$. So there is $r_1, \ldots, r_n \in R$

and $x_1, \ldots, x_n \in K$ with $\sum_{i=1}^n r_i \otimes x_i = x$. Thus we apply ϕ to x:

$$\begin{aligned} y &= \phi(x) \\ &= \phi(\sum_{i=1}^n r_i \otimes x_i) \\ &= \sum_{i=1}^n \phi(r_i \otimes x_i) \\ &= \sum_{i=1}^n \phi(r_i(1 \otimes x_i)) \\ &= \sum_{i=1}^n r_i \phi(1 \otimes x_i) \in RN \end{aligned}$$

So we get that $RN = M$. □

5. Finitely Generated Objects

Definition 5.1. *Let $\mathcal{C}$ be a locally small category that admits filtered colimits of monomorphisms. An object $X \in \mathcal{C}$ is finitely generated, if the representable functor $Hom_{\mathcal{C}}(X, -); \mathcal{C} \to Set$ preserve filtered colimits of monomorphisms.*

Proposition 5.1. *Let $\mathcal{C}$ with initial object 0. Then 0 is finitely generated.*

Proof. Eduardo □

Proposition 5.2. *Let $\mathcal{C}$ be a category with finite coproducts. Then the finite coproduct of finitely generated objects is finitely generated.*

Proof. Let $\mathcal{D}$ be a filtered category, $F : \mathcal{D} \to \mathcal{C}$ a functor and X, Y finitely generated objects, notice that

$$\begin{aligned} colim(Hom_{\mathcal{C}}(X \amalg Y, F)) &\cong colim(Hom_{\mathcal{C}}(X, F) \times Hom_{\mathcal{C}}(Y, F)) \\ &\cong colim(Hom_{\mathcal{C}}(X, F)) \times colim(Hom_{\mathcal{C}}(Y, F)) \\ &\cong Hom_{\mathcal{C}}(X, colim(F)) \times Hom_{\mathcal{C}}(Y, colim(F)) \\ &\cong Hom_{\mathcal{C}}(X \amalg Y, colim(F)) \end{aligned}$$

therefore $X \amalg Y$ is a finitely generated object.

□

6. $K_0(A)$

Definition 6.1. *Let $\mathcal{C}$ be the a monoidal category and A a monoid over $\mathcal{C}$. We define $\mathcal{P}(A)$ as the class of finitely generated monoidal projective left A-modules. We denote by $\mathfrak{P}(A)$ the quotient class of $\mathcal{P}(A)$ modulus the equivalence relation of being isomorphic. We denote by $[P]$ the elements of $\mathfrak{P}(A)$ with $P \in \mathcal{P}(A)$. We will assume that $\mathfrak{P}(A)$ is a set.*

Definition 6.2. *Let $\mathcal{C}$ be the a monoidal category with finite coproducts such that the tensor commutes with finite coproducts. We will call such category a K-monoidal category.*

Proposition 6.1. *Let $\mathcal{C}$ be the a K-monoidal category and A a monoid over $\mathcal{C}$. Then $\mathfrak{P}(A)$ is a commutative monoid with $[P]+[Q] := [P \oplus Q]$ for any $P, Q \in \mathcal{P}(A)$.*

Proof. Let $[P], [Q] \in \mathfrak{P}(A)$, then P and Q are finitely generated $\otimes$-projective left A-modules, it follows from propositions 5.2 and 3.2 that $P \oplus Q$ is a finitely generated $\otimes$-projective left A-module, therefore $[P] + [Q] \in \mathfrak{P}(A)$. Let $[P_1] = [P_2]$ and $[Q_1] = [Q_2]$ be elements in $\mathfrak{P}(A)$. Since $P_1 \cong P_2$ and $Q_1 \cong Q_2$ it follows that $P_1 \oplus Q_1 \cong P_2 \oplus Q_2$, thus $[P_1] + [Q_1] = [P_2] + [Q_2]$. This shows that the sum is well defined.

On the other hand, let $[P], [Q]$ and $[S]$ be elements in $\mathfrak{P}(A)$. Notice that $[P] + ([Q] + [S]) = [P] + [Q \oplus S] = [P \oplus (Q \oplus S)] = [(P \oplus Q) \oplus S] = [P \oplus Q] + [S] = ([P] + [Q]) + [S]$. Moreover, $[P] + [I] = [P \oplus I] = [P]$ for every $[P] \in \mathfrak{P}(A)$. We conclude that $\mathfrak{P}(A)$ is a commutative monoid. □

7. Monoidal change of rings (Part I)

Let $\mathcal{C}$ be a monoidal category, and let (R, μ, η), (S, δ, ε) be monoids in $\mathcal{C}$, with $f : R \to S$ a morphism of monoids. We would like to define a functor

$$\text{S-Mod} \xrightarrow{f^*} \text{R-Mod}\,.$$

Let (M, ρ) be a left S-module. We define $f^*(M, \rho) := (M, \rho^*)$ where $\rho^* = \rho(f \otimes 1_M)$

$$R \otimes M \xrightarrow{f \otimes 1_M} S \otimes M \xrightarrow{\rho} M.$$

Proposition 7.1. *Let (M, ρ) be a left S-module, then $f^*(M, \rho) := (M, \rho^*)$ is a left R-module.*

Proof. Since the following diagrams commute:

$$\begin{tikzcd}
(R \otimes R) \otimes M \arrow[r, "(f \otimes f) \otimes 1_M"] \arrow[d, "\mu \otimes 1_M"'] & (S \otimes S) \otimes M \arrow[d, "\delta \otimes 1_M"] \\
R \otimes M \arrow[r, "f \otimes 1_M"'] & S \otimes M
\end{tikzcd}$$

$$\begin{tikzcd}
(R \otimes R) \otimes M \arrow[r, "\alpha_{R,R,M}"] \arrow[d, "(f \otimes f) \otimes 1_M"'] & R \otimes (R \otimes M) \arrow[d, "f \otimes (f \otimes 1_M)"] \\
(S \otimes S) \otimes M \arrow[r, "\alpha_{S,S,M}"'] & S \otimes (S \otimes M)
\end{tikzcd}$$

we deduce that the following diagram commutes:

$$\begin{tikzcd}
(R \otimes R) \otimes M \arrow[r, "\alpha_{R,R,M}"] \arrow[d, "\mu \otimes 1_M"'] & R \otimes (R \otimes M) \arrow[d, "1_R \otimes \rho^*"] \\
R \otimes M \arrow[r, "\rho^*"'] & M & R \otimes M. \arrow[l, "\rho^*"]
\end{tikzcd}$$

On the other hand, since M is a left S-module and $f : R \to S$ is a morphism of monoids, we deduce that the following diagrams commute:

$$\begin{tikzcd} e \otimes M \arrow[d, "\varepsilon\otimes 1_M"'] \arrow[dr, "\lambda_M^{-1}"] & \\ S \otimes M \arrow[r, "\rho"'] & M \end{tikzcd} \qquad \begin{tikzcd} e \arrow[r, "\eta"] \arrow[dr, "\varepsilon"'] & R \arrow[d, "f"] \\ & S \end{tikzcd}$$

therefore the following diagram commutes:

$$\begin{tikzcd} e \otimes M \arrow[d, "\eta\otimes 1_M"'] \arrow[dr, "\lambda_M^{-1}"] & \\ R \otimes M \arrow[r, "\rho^*"'] & M \end{tikzcd}$$

Thus, $f^*(M,\rho) = (M,\rho^*)$ is a left R-module. □

Let $g : (M,\rho) \to (N,\sigma)$ be a morphism of left S-modules. We define $f^*(g) := g$.

Proposition 7.2. *Let $g : (M,\rho) \to (N,\sigma)$ be a morphism of left S-modules. Then, $g : (M,\rho^*) \to (N,\sigma^*)$ is a morphism of left R-modules.*

Proof. Since the following diagram commutes:

$$\begin{tikzcd} R \otimes M \arrow[r, "1_R\otimes g"] \arrow[d, "f\otimes 1_M"'] & R \otimes N \arrow[d, "f\otimes 1_N"] \\ S \otimes M \arrow[r, "1_S\otimes g"] \arrow[d, "\rho"'] & S \otimes N \arrow[d, "\sigma"] \\ M \arrow[r, "g"'] & N \end{tikzcd}$$

we deduce the following diagram commutes:

$$\begin{tikzcd} R \otimes M \arrow[r, "1_R\otimes g"] \arrow[d, "\rho^*"'] & R \otimes N \arrow[d, "\sigma^*"] \\ M \arrow[r, "g"'] & N. \end{tikzcd}$$

Therefore $g : (M,\rho^*) \to (N,\sigma^*)$ is a morphism of left R-modules. □

Let $\mathcal{C}$ be a monoidal category, and let (R,μ,η), (S,δ,ε) be monoids in $\mathcal{C}$, with $f : R \to S$ a morphism of monoids. We would like to define a functor

$$\text{R-Mod} \xrightarrow{f_!} \text{S-Mod}\,.$$

In order to define the functor, we present an auxiliary construction. Let (M,ρ_l,ρ_r) be a (R,S)-bimodule, and let (N,σ) be a left S-module. Furthermore, suppose that $\mathcal{C}$ admits co-equalizers. Consider the following diagram where (Q,q) is the co-equalizer of the pair of morphisms specified below:

$$M \otimes (S \otimes N) \underset{1_M\otimes\sigma}{\overset{(\rho_r\otimes 1_N)\alpha^{-1}_{M,S,N}}{\rightrightarrows}} M \otimes N \overset{q}{\dashrightarrow} Q.$$

We denote $M \otimes_S N := Q$. Now we look for a morphism $\widehat{\rho_l} : R \otimes (M \otimes_S N) \to M \otimes_S N$ in $\mathcal{C}$ such that $(M \otimes_S N, \widehat{\rho_l})$ is a left R-module. Suppose that the bifunctor $-\otimes- : \mathcal{C}\times\mathcal{C} \to \mathcal{C}$ preserves co-equalizers in each variable, then we have the following commutative diagram:

$$\begin{tikzcd}
R \otimes (M \otimes (S \otimes N)) \arrow[r, shift left, "1_R \otimes ((\rho_r \otimes 1_N)\alpha^{-1}_{M,R,N})"] \arrow[r, shift right, "1_R \otimes (1_M \otimes \sigma)"'] & R \otimes (M \otimes N) \arrow[r, dashed, "1_R \otimes q"] \arrow[d, "(\rho_l \otimes 1_N)\alpha^{-1}_{R,M,N}"] & R \otimes (M \otimes_S N) \arrow[d, dashed, "\widehat{\rho_l}"] \\
& M \otimes N \arrow[r, "q"'] & M \otimes_S N
\end{tikzcd}$$

where the morphism $\widehat{\rho_l} : R \otimes (M \otimes_S N) \to M \otimes_S N$ is obtained by means of the universal property applied to the co-equalizer $(R \otimes (M \otimes_S N), 1_R \otimes q)$.

Proposition 7.3. *Let (M, ρ_l, ρ_r) be a (R, S)-bimodule, and let (N, σ) be a left S-module. Then, $(M \otimes_S N, \widehat{\rho_l})$ is a left R-module.*

Proof. We must verify that the following diagrams commute:

$$\begin{tikzcd}
(R \otimes R) \otimes (M \otimes_S N) \arrow[rr, "\alpha_{R,R,M \otimes_S N}"] \arrow[d, "\mu \otimes 1_{M \otimes_S N}"'] & & R \otimes (R \otimes (M \otimes_S N)) \arrow[d, "1_R \otimes \widehat{\rho_l}"] \\
R \otimes (M \otimes_S N) \arrow[r, "\widehat{\rho_l}"'] & M \otimes_S N & R \otimes (M \otimes_S N) \arrow[l, "\widehat{\rho_l}"]
\end{tikzcd}$$

$$\begin{tikzcd}
e \otimes (M \otimes_S N) \arrow[d, "\eta \otimes 1_{M \otimes_S N}"'] \arrow[dr, "\lambda^{-1}_{M \otimes_S N}"] & \\
R \otimes (M \otimes_S N) \arrow[r, "\widehat{\rho_l}"'] & M \otimes_S N
\end{tikzcd}$$

□

We are now in a position to define our functor at the object level. Let (M, ρ) be a left R-module. We define $f_!(M, \rho) = (S \otimes_R M, \widehat{\delta})$.

Proposition 7.4. *Let (M, ρ) be a left R-module. Then, $f_!(M)$ is a left S-module.*

Proof. Notice that S is an (S, R)-bimodule with right action given by

$$S \otimes R \xrightarrow{1_S \otimes f} S \otimes S \xrightarrow{\delta} S$$

and left action given by

$$S \otimes S \xrightarrow{\delta} S.$$

Therefore by proposition 7.3 we conclude that $f_!(M)$ is a left S-module. □

We now study the assignment at the level of morphisms. Let $g : (M, \rho) \to (N, \sigma)$ a morphism of left R-modules. We have the following commutative diagram:

$$\begin{tikzcd}
S \otimes (S \otimes M) \arrow[r, shift left, "(\delta \otimes 1_M)\alpha^{-1}_{S,S,M}"] \arrow[r, shift right, "1_S \otimes \rho"'] & S \otimes M \arrow[r, "q"] \arrow[d, "1_S \otimes g"] & S \otimes_R M \arrow[d, dashed, "1_S \otimes_R g"] \\
& S \otimes N \arrow[r, "q'"'] & S \otimes_R N
\end{tikzcd}$$

where the morphism $1_S \otimes_R g : S \otimes_R M \to S \otimes_R N$ is obteined by means of the universal property applied to the co-equalizer $(S \otimes_R M, q)$. We define $f_!(g) := 1_S \otimes_R g$.

Proposition 7.5. *Let $g : (M, \rho) \to (N, \sigma)$ a morphism of left R-modules. Then $f_!(g) : f_!(M) \to f_!(N)$ is morphism of left S-modules.*

8. Monoidal change of rings (Part II)

Let $\mathcal{C}$ be a monoidal category, R, S monoids in $\mathcal{C}$, and let $f\colon R \to S$ be a morphism of monoids. Throughout this section it is assumed that the bifunctor $- \otimes - : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ preserves coequalizers in both variables.

The goal of this section is to obtain the following adjunction

$$\begin{tikzcd} \text{R-Mod} \arrow[r, bend left, "f_!"] \arrow[r, phantom, "\perp"] & \text{S-Mod}\,. \arrow[l, bend left, "f^*"] \end{tikzcd}$$

In order to obtain the adjunction, it will be verified that the triangular identities are satisfied. We begin by studying the definition of the counit.

It must be the case that $\varepsilon\colon f_! f^* \to 1_S$, where 1_S denotes the identity functor $1_{\text{S-Mod}}\colon \text{S-Mod} \to \text{S-Mod}$. In particular, for $M \in \text{S-Mod}$ one must have $\varepsilon_M : S \otimes_R M^* \to M$ a morphism of S-modules. This morphism will be obtained by means of the universal property of the S-module $S \otimes_R M^*$, as explained below.

Since M is an S-module, there exists a left action $\rho : S \otimes M \to M$. By the universal property of the coequalizer $S \otimes_R M$, it suffices to check that $\rho a = \rho b$, where $a = (\mu_S \otimes 1_{M^*})\alpha^{-1}_{S,R,M^*}$ and $b = 1_S \otimes \rho^*$, since this guarantees the existence of a morphism $\varepsilon_M : S \otimes_R M^* \to M$ such that the following diagram commutes:

$$\begin{tikzcd} S \otimes (R \otimes M^*) \arrow[r, shift left, "a"] \arrow[r, shift right, "b"'] & S \otimes M^* \arrow[r, "q"] \arrow[d, "\rho"'] & S \otimes_R M^* \arrow[dl, dashed, "\varepsilon_M"] \\ & M. & \end{tikzcd}$$

From the commutativity of the following diagram:

$$\begin{tikzcd} (S \otimes R) \otimes M \arrow[r, "\alpha_{S,R,M}"] \arrow[d, "(1_S \otimes f) \otimes 1_M"'] & S \otimes (R \otimes M) \arrow[d, "1_S \otimes (f \otimes 1_M)"] \\ (S \otimes S) \otimes M \arrow[r, "\alpha_{S,S,M}"] \arrow[d, "\mu_S \otimes 1_M"'] & S \otimes (S \otimes M) \arrow[d, "1_S \otimes \rho"] \\ S \otimes M \arrow[r, "\rho"'] & M & S \otimes M \arrow[l, "\rho"] \end{tikzcd}$$

it follows that $\rho a = \rho b$.

Proposition 8.1. *Let M be an S-module, then $\varepsilon_M : f_! f^*(M) \to M$ is a morphism of S-modules.*

Proof. Since M is an S-module, there exists a left action $\rho : S \otimes M \to M$. One must verify that the following diagram commutes:

$$\begin{tikzcd} S \otimes (S \otimes_R M^*) \arrow[r, "1_S \otimes \varepsilon_M"] \arrow[d, "\tilde{\mu_S}"'] & S \otimes M \arrow[d, "\rho"] \\ S \otimes_R M^* \arrow[r, "\varepsilon_M"'] & M. \end{tikzcd}$$

From the definition of the action $\tilde{\mu_S}$ and the morphism ε_M, we obtain the following commutative diagram:

$$\begin{tikzcd}
S \otimes (S \otimes (R \otimes M^*)) \arrow[r, shift left, "1_S \otimes a"] \arrow[r, shift right, "1_S \otimes b"'] & S \otimes (S \otimes M^*) \arrow[r, "1_S \otimes q"] \arrow[d, "\alpha^{-1}_{S,S,M}"] & S \otimes (S \otimes_R M^*) \arrow[dd, "\tilde{\mu_S}"] \\
& (S \otimes S) \otimes M^* \arrow[d, "\mu_S \otimes 1_M"] & \\
S \otimes (R \otimes M^*) \arrow[r, shift left, "a"] \arrow[r, shift right, "b"'] & S \otimes M^* \arrow[r, "q"'] \arrow[d, "\rho"] & S \otimes_R M^* \arrow[dl, "\varepsilon_M"] \\
& M. &
\end{tikzcd}$$

Observe that

$$\begin{aligned}
\varepsilon_M \tilde{\mu_S}(1_S \otimes q) =& \varepsilon_M q(\mu_S \otimes 1_M)\alpha^{-1}_{S,S,M} \\
=& \rho(\mu_S \otimes 1_M)\alpha^{-1}_{S,S,M} \\
=& \rho(1_S \otimes \rho) \\
=& \rho(1_S \otimes (\varepsilon_M q)) \\
=& \rho(1_S \otimes \varepsilon_M)(1_S \otimes q).
\end{aligned}$$

Therefore $\varepsilon_M \tilde{\mu_S} = \rho(1_S \otimes \varepsilon_M)$, since the morphism $1_S \otimes q$ is an epimorphism by virtue of the fact that the bifunctor $- \otimes - : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ preserves coequalizers in the second variable. □

Proposition 8.2. *The assignment $\varepsilon : f_! f^* \to 1_S$ is a natural transformation.*

Proof. Let $g : M \to N$ be a morphism of left S-modules. One must verify that the following diagram commutes:

$$\begin{tikzcd}
S \otimes_R M^* \arrow[r, "1_S \otimes_R g^*"] \arrow[d, "\varepsilon_M"'] & S \otimes_R N^* \arrow[d, "\varepsilon_N"] \\
M \arrow[r, "g"'] & N.
\end{tikzcd}$$

Since M, N are left S-modules, there exist left actions $\rho : S \otimes M \to M$ and $\sigma : S \otimes N \to N$. Then, by definition of the morphisms $1_S \otimes_R g^*$ and ε_M, ε_N, we obtain the following diagram:

$$\begin{tikzcd}
S \otimes M^* \arrow[r, "q"] \arrow[d, "\rho"'] \arrow[dd, bend right, "1_S \otimes g^*"'] & S \otimes_R M^* \arrow[dl, "\varepsilon_M"] \arrow[ddd, "1_S \otimes_R g^*"] \\
M \arrow[d, "g"'] & \\
N & \\
S \otimes N^* \arrow[r, "q'"'] \arrow[u, "\sigma"] & S \otimes_R N^* \arrow[ul, "\varepsilon_N"']
\end{tikzcd}$$

Notice that

$$\begin{aligned}\varepsilon_N(1_S \otimes_R g^*)q =& \varepsilon_N q'(1_S \otimes g^*)\\ =& \sigma(1_S \otimes g^*)\\ =& g\rho\\ =& g\varepsilon_M q.\end{aligned}$$

Therefore $\varepsilon_N(1_S \otimes_R g^*) = g\varepsilon_M$, since q is an epimorphism.

□

We continue by studying the definition of the unit.

It must be the case that $\delta : 1_R \to f^* f_!$, where 1_R denotes the identity functor $1_{\text{R-Mod}} : \text{R-Mod} \to \text{R-Mod}$. In particular, for $M \in \text{R-Mod}$ one must have $\delta_M : M \to (S \otimes_R M)^*$ a morphism of R-modules. This morphism is defined as $\delta_M := q(\eta_S \otimes 1_M)\lambda_M^{-1}$. The following diagram illustrates the situation:

$$\begin{tikzcd} S \otimes (R \otimes M) \arrow[r, shift left, "a"] \arrow[r, shift right, "b"'] & S \otimes M \arrow[r, "q"] & S \otimes_R M \\ & 1 \otimes M \arrow[u, "\eta_S \otimes 1_M"] & M. \arrow[l, "\lambda_M^{-1}"] \arrow[u, dashed, "\delta_M"'] \end{tikzcd}$$

Proposition 8.3. *Let M be an R-module, then $\delta_M : M \to (S \otimes_R M)^*$ is a morphism of R-modules.*

Proposition 8.4. *The assignment $\delta : 1_R \to f^* f_!$ is a natural transformation.*

Proposition 8.5. *Let $f : R \to S$ be a morphism of monoids. The functors*

$$\begin{tikzcd} \text{R-Mod} \arrow[r, bend left, "f_!"] & \text{S-Mod}\,. \arrow[l, bend left, "f^*"] \end{tikzcd}$$

are adjoint, with $f_!$ the left adjoint functor.

Proof. It suffices to verify that the triangular identities are satisfied. First, we show that the following diagram commutes:

$$\begin{tikzcd} f_! \arrow[rr, "f_!(\delta)"] \arrow[dr, "1_{f_!}"'] & & f_! f^* f_! \arrow[dl, "\varepsilon(f_!)"] \\ & f_! & \end{tikzcd}$$

In particular, for M an R-module, one must verify that the following diagram commutes:

$$\begin{tikzcd} S \otimes_R M \arrow[rr, "1_S \otimes_R \delta_M"] \arrow[dr, "1_{S \otimes_R M}"'] & & S \otimes_R (S \otimes_R M)^* \arrow[dl, "\varepsilon_{S \otimes_R M}"] \\ & S \otimes_R M. & \end{tikzcd}$$

By definition of the morphisms $1_S \otimes_R \delta_M$ and $\varepsilon_{S\otimes_R M}$, we obtain the following commutative diagram:

$$\begin{tikzcd}
S\otimes(R\otimes M) \arrow[r, shift left, "a"] \arrow[r, shift right, "b"'] & S\otimes M \arrow[r, "q"] \arrow[d, "1_S\otimes\delta_M"] & S\otimes_R M \arrow[d, "1_S\otimes_R\delta_M"] \\
S\otimes(R\otimes(S\otimes_R M)^*) \arrow[r, shift left, "r"] \arrow[r, shift right, "s"'] & S\otimes(S\otimes_R M)^* \arrow[r, "t"] \arrow[d, "\tilde{\mu_S}"] & S\otimes_R(S\otimes_R M)^* \arrow[dl, "\varepsilon_{S\otimes_R M}"] \\
 & S\otimes_R M. &
\end{tikzcd}$$

Notice that

$$\begin{aligned}
\varepsilon_{S\otimes_R M}(1_S\otimes_R\delta_M)q =& \varepsilon_{S\otimes_R M}t(1_S\otimes\delta_M)\\
=& \tilde{\mu_S}(1_S\otimes\delta_M).
\end{aligned}$$

Then, it suffices to show that $\tilde{\mu_S}(1_S\otimes\delta_M)=q$. Since q is an epimorphism. By definition of δ_M, the following holds:

$$\begin{aligned}
\tilde{\mu_S}(1_S\otimes\delta_M) =& \tilde{\mu_S}(1_S\otimes(q(\eta_S\otimes 1_M)\lambda_M^{-1}))\\
=& \tilde{\mu_S}(1_S\otimes q)(1_S\otimes(\eta_S\otimes 1_M))(1_S\otimes\lambda_M^{-1}).
\end{aligned}$$

By definition of $\tilde{\mu_S}$, we obtain the following commutative diagram:

$$\begin{tikzcd}
S\otimes(S\otimes M) \arrow[r, "1_S\otimes q"] \arrow[d, "\alpha^{-1}_{S,S,M}"'] & S\otimes(S\otimes_R M) \arrow[dd, dashed, "\tilde{\mu_S}"] \\
(S\otimes S)\otimes M \arrow[d, "\mu_S\otimes 1_M"'] & \\
S\otimes M \arrow[r, "q"'] & S\otimes_R M.
\end{tikzcd}$$

Then

$$\tilde{\mu_S}(1_S\otimes\delta_M) = q(\mu_S\otimes 1_M)\alpha^{-1}_{S,S,M}(1_S\otimes(\eta_S\otimes 1_M))(1_S\otimes\lambda_M^{-1}).$$

On the other hand, since $\alpha_{S,-,M}$ is natural in the second variable, we obtain the following commutative diagram:

$$\begin{tikzcd}
S\otimes(1\otimes M) \arrow[r, "\alpha^{-1}_{S,1,M}"] \arrow[d, "1_S\otimes(\eta_S\otimes 1_M)"'] & (S\otimes 1)\otimes M \arrow[d, "(1_S\otimes\eta_S)\otimes 1_M"] \\
S\otimes(S\otimes M) \arrow[r, "\alpha^{-1}_{S,S,M}"'] & (S\otimes S)\otimes M.
\end{tikzcd}$$

Therefore

$$\begin{aligned}
\tilde{\mu_S}(1_S\otimes\delta_M) =& q(\mu_S\otimes 1_M)((1_S\otimes\eta_S)\otimes 1_M)\alpha^{-1}_{S,1,M}(1_S\otimes\lambda_M^{-1})\\
=& q(\mu_S(1_S\otimes\eta_S)\otimes 1_M)\alpha^{-1}_{S,1,M}(1_S\otimes\lambda_M^{-1}).
\end{aligned}$$

Since S is a monoid in $\mathcal{C}$, we obtain the following commutative diagram:

$$\begin{tikzcd} 1 \otimes S \arrow[rr, "1_S \otimes \eta_S"] \arrow[dr, "\rho_S"'] & & S \otimes S \arrow[dl, "\mu_S"] \\ & S. & \end{tikzcd}$$

Thus

$$\tilde{\mu_S}(1_S \otimes \delta_M) = q(\rho_S \otimes 1_M)\alpha^{-1}_{S,1,M}(1_S \otimes \lambda^{-1}_M).$$

Since $\mathcal{C}$ is a monoidal category, we obtain the following commutative diagram:

$$\begin{tikzcd} (S \otimes 1) \otimes M \arrow[rr, "\alpha_{S,1,mM}"] \arrow[dr, "\rho_S \otimes 1_M"'] & & S \otimes (1 \otimes M) \arrow[dl, "1_S \otimes \lambda_M"] \\ & S \otimes M. & \end{tikzcd}$$

Thus

$$\tilde{\mu_S}(1_S \otimes \delta_M) = q.$$

Finally, we verify that the following diagram commutes:

$$\begin{tikzcd} f^* \arrow[rr, "\delta_{f^*}"] \arrow[dr, "1_{f^*}"'] & & f^* f_! f^* \arrow[dl, "f^*(\varepsilon)"] \\ & f^*. & \end{tikzcd}$$

In particular, for M an S-module, one must verify that the following diagram commutes:

$$\begin{tikzcd} M^* \arrow[rr, "\delta_{f^*}"] \arrow[dr, "1_{M^*}"'] & & (S \otimes_R M^*)^* \arrow[dl, "(\varepsilon_M)^*"] \\ & M^*. & \end{tikzcd}$$

By definition of the morphisms δ_{M^*} and $(\varepsilon_M)^*$, we obtain the following commutative diagram:

$$\begin{tikzcd} & 1 \otimes M^* \arrow[d, "\eta_S \otimes 1_{M^*}"] & M^* \arrow[l, "\lambda^{-1}_{M^*}"'] \arrow[d, "\delta_{M^*}"] \\ S \otimes (R \otimes M^*) \arrow[r, shift left, "a"] \arrow[r, shift right, "b"'] & S \otimes M^* \arrow[r, "q"] \arrow[d, "\rho"'] & S \otimes_R M^* \arrow[dl, "\varepsilon_M"] \\ & M & \end{tikzcd}$$

Notice that

$$\begin{aligned} \varepsilon_M \delta_{M^*} &= \varepsilon_M q(\eta_S \otimes 1_{M^*})\lambda^{-1}_{M^*} \\ &= \rho(\eta_S \otimes 1_{M^*})\lambda^{-1}_{M^*}. \end{aligned}$$

Since M is an S-module, we obtain the following commutative diagram:

$$\begin{tikzcd} 1 \otimes M \arrow[r, "\lambda_M"] \arrow[d, "\eta_S \otimes 1_M"'] & M \arrow[d, "1_M"] \\ S \otimes M \arrow[r, "\rho"'] & M. \end{tikzcd}$$

Therefore

$$(\varepsilon_M)^* \delta_{M^*} = 1_{M^*}.$$

We conclude that $(f_!, f^*) : \text{R-Mod} \to \text{S-Mod}$ is an adjoint pair.

□

9. The functor $K_0^{\otimes}$

Let $\mathcal{C}$ be a monoidal category. We denote by $\mathrm{Mon}(\mathcal{C})$ the category of monoids in $\mathcal{C}$. The goal of this section is to define a functor

$$K_0^{\otimes} : \mathrm{Mon}(\mathcal{C}) \to \mathrm{Ab}\,.$$

To achieve this goal, we will make use of the Grothendieck completion functor. First, a construction at the level of commutative monoids is required.

Let R be a monoid in $\mathcal{C}$. We define $\mathbf{P}(R)$ as the set of equivalence classes of finitely generated $\otimes$-projective R-modules.

Proposition 9.1. *Let R be a monoid in $\mathcal{C}$ and $[P], [Q] \in \mathbf{P}(R)$. We define*

$$[P] + [Q] := [P \oplus Q].$$

Then $(\mathbf{P}(R), +)$ is a commutative monoid.

Proof. Let $[P], [Q] \in \mathbf{P}(R)$. Since the coproduct of $\otimes$-projective objects is $\otimes$-projective and the coproduct of finitely generated objects is finitely generated, it follows that $[P \oplus Q] \in \mathbf{P}(R)$, that is, $[P] + [Q] \in \mathbf{P}(R)$.

Let $[P_1] = [P_2], [Q_1] = [Q_2] \in \mathbf{P}(R)$. Then $P_1 \cong P_2$ and $Q_1 \cong Q_2$ in $\mathcal{C}$, hence $P_1 \oplus Q_1 \cong P_2 \oplus Q_2$ in $\mathcal{C}$. Therefore,

$$[P_1] + [Q_1] = [P_1 \oplus Q_1] = [P_2 \oplus Q_2] = [P_2] + [Q_2].$$

That is, the operation is well defined.

Let $[P_1], [P_2], [P_3] \in \mathbf{P}(R)$. Observe that

$$[P_1]+([P_2]+[P_3]) = [P_1]+[P_2 \oplus P_3] = [P_1 \oplus (P_2 \oplus P_3)] = [(P_1 \oplus P_2) \oplus P_3] = [P_1 \oplus P_2]+[P_3] = ([P_1]+[P_2])+[P_3].$$

Let $[P] \in \mathbf{P}(R)$. Consider 0 to be the initial object of the monoidal category $\mathcal{C}$. Since 0 is $\otimes$-projective and finitely generated, it follows that $[0] \in \mathbf{P}(R)$. Moreover,

$$[P] + [0] = [P \oplus 0] = [P] = [0 \oplus P] = [0] + [P].$$

Having verified all the conditions, we conclude that $(\mathbf{P}(R), +)$ is a commutative monoid. □

Let $f : R \to S$ be a morphism of monoids in $\mathcal{C}$. We define $\mathbf{P}(f) : \mathbf{P}(R) \to \mathbf{P}(S)$ by $\mathbf{P}(f)([Q]) := [f_!(Q)] = [S \otimes_R Q]$.

Lemma 9.1. *Let $(L, R) : \mathcal{C} \to \mathcal{D}$ be an adjoint pair and let $P \in \mathcal{D}$. Suppose there exists $X \in \mathcal{C}$ such that P is a retract of $L(X)$; then P is LR-projective.*

Proof. By definition, it suffices to show that $\varepsilon_P : LR(P) \to P$ is an epimorphism that splits in $\mathcal{D}$. Since P is a retract of $L(X)$, we obtain the following commutative diagram:

$$\begin{tikzcd} & P \arrow[dl, "\beta"'] \arrow[d, "1_P"] \\ L(X) \arrow[r, "\alpha"'] & P \end{tikzcd}$$

Considering the natural transformation $\varepsilon : LR \to 1_{\mathcal{D}}$ and the morphism $\alpha : L(X) \to P$, we obtain the following commutative diagram:

$$\begin{tikzcd} LR(L(X)) \arrow[r, "LR(\alpha)"] \arrow[d, "\varepsilon_{L(X)}"'] & LR(P) \arrow[d, "\varepsilon_P"] \\ L(X) \arrow[r, "\alpha"'] & P \end{tikzcd}$$

That is, $\varepsilon_P LR(\alpha) = \alpha\varepsilon_{L(X)}$. Since $L(X)$ is an LR-projective object, $\varepsilon_{L(X)}$ is an epimorphism that splits, as does α. Therefore, ε_P is an epimorphism that splits in $\mathcal{D}$. □

Lemma 9.2. *Let $f : R \to S$ be a morphism of monoids in $\mathcal{C}$ and let $P \in$ R-Mod. If P is $\otimes$-projective, then $f_!(P) = S \otimes_R P$ is $\otimes$-projective.*

Proof. By hypothesis, $P \in$ R-Mod is $\otimes$-projective, that is, the counit $\varepsilon_P^R : F^R U^R(P) \to P$ is an epimorphism that splits in R-Mod. Therefore, $f_!(\varepsilon_P^R) : f_!(F^R U^R(P)) \to f_!(P)$ is an epimorphism that splits in S-Mod. On the other hand, we have

$$f_!(F^R U^R(P)) = S \otimes_R (R \otimes U^R(P)) \cong (S \otimes_R R) \otimes U^R(P) = S \otimes U^R(P) = F^S(U^R(P)).$$

Therefore, $f_!(P)$ is a retract of $F^S(U^R(P))$. It follows from Lemma 9.1 that $f_!(P)$ is a $\otimes$-projective S-module. □

Lemma 9.3. *Let $f : R \to S$ be a morphism of monoids in $\mathcal{C}$. The functor $f^* :$* S-Mod $\to$ R-Mod *preserves coproducts.*

Proof. Let $\{(M_i, \rho_i)\}_{i \in I}$ be a collection of S-modules. Since

$$f^*\left(\coprod_{i \in I} M_i\right) = \coprod_{i \in I} M_i = \coprod_{i \in I} f^*(M_i),$$

it remains to verify that their actions as R-modules coincide. By definition of the action given to the coproduct $\coprod_{i \in I} M_i$ as an S-module and by the definition of the functor $f^* :$ S-Mod $\to$ R-Mod, we obtain the following commutative diagrams:

$$\begin{tikzcd} R \otimes M_j \arrow[r, "f \otimes 1_{M_j}"] \arrow[d, "1_R \otimes q_j"'] & S \otimes M_j \arrow[r, "\rho_j"] \arrow[d, "1_S \otimes q_j"'] & M_j \arrow[d, "q_j"] \\ R \otimes (\coprod_{i \in I} M_i) \arrow[r, "f \otimes 1_{\coprod_{i \in I} M_i}"'] & S \otimes (\coprod_{i \in I} M_i) \arrow[r, dashed, "\rho_S"'] & \coprod_{i \in I} M_i \end{tikzcd}$$

$$\begin{tikzcd}
R \otimes M_j \arrow[r, "f \otimes 1_{M_j}"] \arrow[d, "1_R \otimes q_j"'] & S \otimes M_j \arrow[r, "\rho_j"] & M_j \arrow[d, "q_j"] \\
R \otimes (\coprod_{i \in I} M_i) \arrow[rr, dashed, "\rho_R"'] & & \coprod_{i \in I} M_i
\end{tikzcd}$$

Notice that

$$\begin{aligned}
\rho_S(f \otimes 1_{\coprod_{i \in I} M_i})(1_R \otimes q_j) =& \rho_S(1_S \otimes q_j)(f \otimes 1_{M_j}) \\
=& q_j \rho_j (f \otimes 1_{M_j}).
\end{aligned}$$

Since ρ_R is the unique morphism with this property, it follows that $\rho_R = \rho_S(f \otimes 1_{\coprod_{i \in I} M_i})$. Therefore $(f^*(\coprod_{i \in I} M_i), \rho_S(f \otimes 1_{M_j})) = (\coprod_{i \in I} f^*(M_i), \rho_R)$ as R-modules.

□

Lemma 9.4. *Let $f : R \to S$ be a morphism of monoids in $\mathcal{C}$. The functor f^* :* S-Mod $\to$ R-Mod *preserves coequalizers.*

Proof. Let $g, h : (M, \rho_M) \to (N, \rho_N)$ be morphisms of S-modules. Consider the coequalizer of g and h in S-Mod:

$$\begin{tikzcd}
M \arrow[r, shift left, "g"] \arrow[r, shift right, "h"'] & N \arrow[r, dashed, "q"] & E
\end{tikzcd}$$

Applying the functor f^* : S-Mod $\to$ R-Mod yields the following situation:

$$\begin{tikzcd}
f^*(M) \arrow[r, shift left, "f^*(g)"] \arrow[r, shift right, "f^*(h)"'] & f^*(N) \arrow[r, "e"] \arrow[dr, "f^*(q)"'] & Q \\
& & f^*(E)
\end{tikzcd}$$

where (Q, e) is the coequalizer of $f^*(g)$ and $f^*(h)$. By the construction of the action for the respective coequalizers, the following commutative diagrams hold:

$$\begin{tikzcd}
R \otimes M \arrow[r, shift left, "1_R \otimes g"] \arrow[r, shift right, "1_R \otimes h"'] & R \otimes N \arrow[r, "1_R \otimes q"] \arrow[d, "f \otimes 1_N"'] & R \otimes E \arrow[d, dashed, "\sigma"] \\
& S \otimes N \arrow[d, "\rho_N"'] & \\
& N \arrow[r, "q"'] & E
\end{tikzcd}$$

$$\begin{tikzcd}
& R \otimes N \arrow[r, "1_R \otimes q"] \arrow[d, "f \otimes 1_N"'] & R \otimes E \arrow[d, "f \otimes 1_E"] \\
S \otimes M \arrow[r, shift left, "1_S \otimes g"] \arrow[r, shift right, "1_S \otimes h"'] & S \otimes N \arrow[r, "1_S \otimes q"] \arrow[d, "\rho_N"'] & S \otimes E \arrow[d, dashed, "\hat{\rho_N}"] \\
& N \arrow[r, "q"'] & E.
\end{tikzcd}$$

Since $f^*(E) = E = Q$ it suffices to see that their actions as R-modules coincide. Observe that:

$$\begin{aligned}\hat{\rho_N}(f \otimes 1_E)(1_R \otimes q) =& \hat{\rho_N}(1_S \otimes q)(f \otimes 1_N)\\ =& q\rho_N(f \otimes 1_N).\end{aligned}$$

Since σ is the unique R-module morphism that satisfies making the first diagram commutative, it follows that $\sigma = \hat{\rho_N}(f \otimes 1_E)$. Therefore $f^* : \text{S-Mod} \to \text{R-Mod}$ preserves coequalizers. □

Corollary 9.1. *Let $f : R \to S$ be a morphism of monoids in $\mathcal{C}$. The functor $f^* : \text{S-Mod} \to \text{R-Mod}$ preserves colimits.*

Proof. It follows from Lemmas 9.3 and 9.4. □

Lemma 9.5. *Let $f : R \to S$ be a morphism of monoids in $\mathcal{C}$ and $P \in \text{R-Mod}$. If P is finitely generated, then $f_!(P) = S \otimes_R P$ is finitely generated.*

Proof.

$$\begin{aligned}Hom_S(f_!(P), \varinjlim_{i\in I}(M_i)) =& Hom_R(P, f^*(\varinjlim_{i\in I}(M_i)))\\ =& Hom_R(P, \varinjlim_{i\in I} f^*(M_i))\\ =& \varinjlim_{i\in I} Hom_R(P, f^*(M_i))\\ =& \varinjlim_{i\in I} Hom_S(f_!(P), M_i).\end{aligned}$$

Where the second equality follows from Corollary 9.1. □

Lemma 9.6. *Let $f : R \to S$ be a morphism of monoids in $\mathcal{C}$, the functor $f_! : \text{R-Mod} \to \text{S-Mod}$ preserves coproducts and zero object.*

Proof. □

Proposition 9.2. *Let $f : R \to S$ be a morphism of monoids in $\mathcal{C}$, then $\mathbf{P}(f) : \mathbf{P}(R) \to \mathbf{P}(S)$ is a morphism of commutative monoids.*

Proof. By Lemmas 9.2 and 9.5 it follows that the assignment $\mathbf{P}(f) : \mathbf{P}(R) \to \mathbf{P}(S)$ is well-defined.

Let $[P], [Q] \in \mathbf{P}(R)$. It follows from Lemma 9.6 that $\mathbf{P}(f)([P]+[Q]) = \mathbf{P}(f)([P \oplus Q]) = [f_!(P \oplus Q)] = [f_!(P) \oplus f_!(Q)] = [f_!(P)] + [f_!(Q)] = \mathbf{P}(f)([P]) + \mathbf{P}(f)([Q])$. Moreover, by Lemma 9.6 it also follows that $\mathbf{P}(f)([0_R]) = [f_!(0_R)] = [0_S]$. Therefore, $\mathbf{P}(f) : \mathbf{P}(R) \to \mathbf{P}(S)$ is a morphism of commutative monoids.

□

Lemma 9.7.

Proof. □

Proposition 9.3. *Let $f : R \to S$ and $g : S \to T$ be morphisms of monoids in $\mathcal{C}$, then $\mathbf{P}(gf) = \mathbf{P}(g)\mathbf{P}(f)$.*

Proof. Let $[P] \in \mathbf{P}(R)$. It follows from Lemma 9.7 that $\mathbf{P}(g)\mathbf{P}(f)([P]) = \mathbf{P}(g)([f_!(P)]) = \mathbf{P}(g)([S \otimes_R P]) = [g_!(S \otimes_R P)] = [T \otimes_S (S \otimes_R P)] = [T \otimes_R P] = [(gf)_!(P)] = \mathbf{P}(gf)([P])$. Therefore, $\mathbf{P}(gf) = \mathbf{P}(g)\mathbf{P}(f)$. □

Lemma 9.8. *Let R be a monoid in $\mathcal{C}$. For every $M \in$ R-Mod it follows that $R \otimes_R M \cong M$.*

Proof. □

Proposition 9.4. *Let R be a monoid in $\mathcal{C}$, then $\mathbf{P}(1_R) = 1_{\mathbf{P}(R)}$.*

Proof. Let $[P] \in \mathbf{P}(R)$. It follows from Lemma 9.8 that $\mathbf{P}(1_R)([P]) = [(1_R)_!(P)] = [R \otimes_R P] = [P] = 1_{\mathbf{P}(R)}([P])$. □

Corollary 9.2. *Let $\mathcal{C}$ be a monoidal category, we have a functor $\mathbf{P} : Mon(\mathcal{C}) \to CMon$.*

Proof. It follows directly from Propositions 9.2, 9.3 and 9.4. □

Definition 9.1.

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